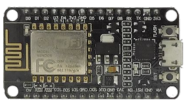
SENSOR SENSEI

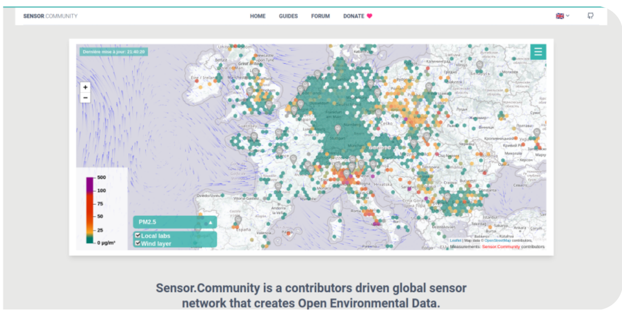
< MASTER OF SOCKETS />

SENSOR SENSEI

The **sensor.community** project aims to pool a civic effort to measure air quality at a global level. Their website provides all the explanations for making your own sensor.

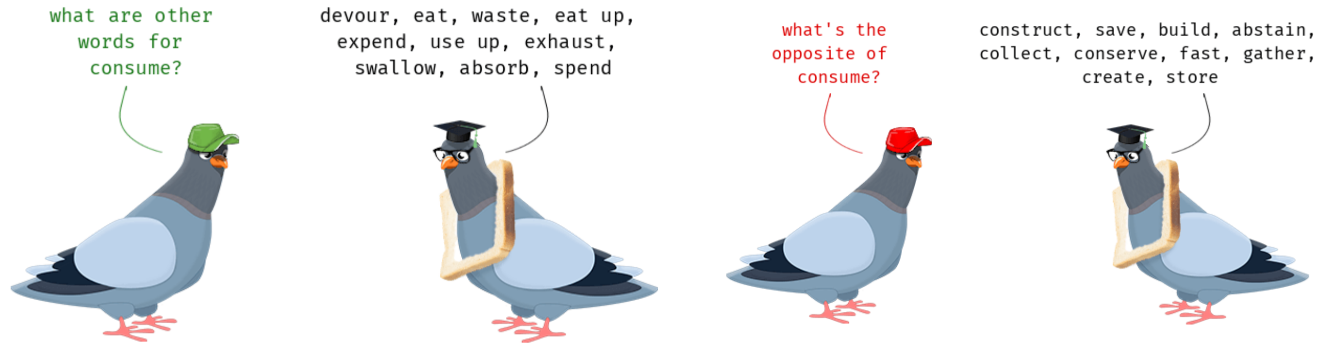
But, one of the limitations of this sensor *which has given rise to several discussions among users* is the need to connect to a wifi access point.





If finding a wifi hotspot is often possible in an urban environment, it is different in rural areas. In addition, in an environmental approach, the use of wifi may be called into question, given its high energy consumption ...

There are transfer technologies that are much more suitable for this kind of use, much less energy-consuming and with a range much greater than wifi, in particular LoRa technology.



Discussions regarding this technology were aimed at adding the ability to connect the sensor to TheThingsNetworks, a global LoRa gateway network. In this implementation, LoRa completely replaces the WiFi.

You must carry out a different implementation of LoRa communication, in the form of a local network, connected via WiFi using a single and dedicated gateway. You must be able to deploy multiple sensors within a radius of a few kilometers around a WiFi access.

Design a sensor compatible with the sensor.community project, adding the possibility of using the LoRa protocol for ”local” long-distance communication and transmitting them via a LoRa > Wifi gateway. Your firmware **must** keep all the features currently existing in the code, like:

- ✓ Wifi access point for configuration and monitoring ;
- ✓ compatibility with the same list of sensors.

The cards provided for the project are based on an ESP32. The compatibility of the current firmware with the ESP32 is still experimental (beta branch).

Depending on your preference, you can base your work on the existing firmware, by improving ESP32 support and then adding the LoRa functionalities, or else writing new firmware, it will then be necessary to rebuild all the existing functionalities.

Delivery

You're expected to provide:

- ✓ A listing of the features of the current firmware ;
- ✓ A remote sensor made up of at least:
  - an autonomous part (battery), which carries out the measurements and transmits them via the LoRa protocol to ... ;
  - another part (gateway) which is responsible for transmitting the message from the sensor (s) to the sensor.community server ;
- ✓ New firmware retaining the functionality of the current one ;
- ✓ A solution that minimizes energy consumption and maximizes the autonomy of the autonomous part ;
- ✓ Documentation (crafting, firmware, user, ...) to facilitate project sharing with the community. Among them, a document details your choice of implementation compared to the current firmware and feedback ;
- ✓ Visualizations (from scratch or from lib) to facilitate the analysis of your data and/or others.

> ⚠ If you're facing some hardware compatibility issues, you are allowed to: hard-code the GPS coordinates from the client, to transmit to sensor.community not use the microphone as a remote sensor.

Bonus

You can improve this project in many ways, including:

- ✓ contribute to sensor.community project on Github ;
- ✓ data visualization and monitoring ;
- ✓ join a campaign ;
- ✓ ...

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